

Making Figures Count

Effective scientific visualisation for publications and grant applications

Aron Hill
Cognitive Neuroscience Unit



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About this talk

WHAT WE'LL COVER

- Follow one *synthetic* dataset from a first draft figure through to a manuscript-worthy figure.
- See some basic code behind each change.
- Explore how different plots alter how the same data can be presented

SUPPLIED MATERIALS

- An illustrated figure pipeline with explanations, code and outputs.
- The data, a starter script and complete R scripts to adapt.

We'll focus on the figure decisions today. The supplied code is your reference afterwards.



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Tools used in this talk

R

The language that reads the data and runs the analysis.

ggplot2

Popular R package that builds figures by adding layers.

Quarto

Bring everything together. Useful for combining code, figures and explanations.



ggplot2

The same plotting principals apply whatever software you decide to use!



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Guide, code and data

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Making Figures Count

Aron Hill — Research Incubator talk.

A guide to making figures in R and ggplot2, using one fictional caffeine and vigilance dataset. All 80 observations are simulated.

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About

Materials for Making Figures Count: a practical guide to creating figures in R and ggplot2, with synthetic data, reusable code and presentation slides.

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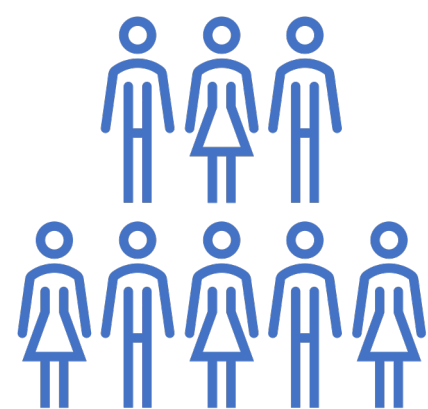
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The example study and dataset

80 simulated participants, randomly allocated to two groups

Placebo

40 participants



Caffeine 200 mg

40 participants



Sustained-attention task

Vigilance score: 0–100. Higher scores indicate better performance.

OK, so you have analysed the data and now would like to plot the key finding of vigilance scores between the active and placebo groups...

caffeine_attention.csv

caffeine_attention			
id	group	metaboliser	vigilance
C18	Caffeine 200 mg	Fast	67.2
C09	Caffeine 200 mg	Fast	77.8
C19	Caffeine 200 mg	Fast	73.8
C37	Caffeine 200 mg	Slow	62.7
C34	Caffeine 200 mg	Slow	44.9
C38	Caffeine 200 mg	Slow	54.1
P14	Placebo	NA	58.2
C13	Caffeine 200 mg	Fast	67.4
C40	Caffeine 200 mg	Slow	54.7
P38	Placebo	NA	52.9

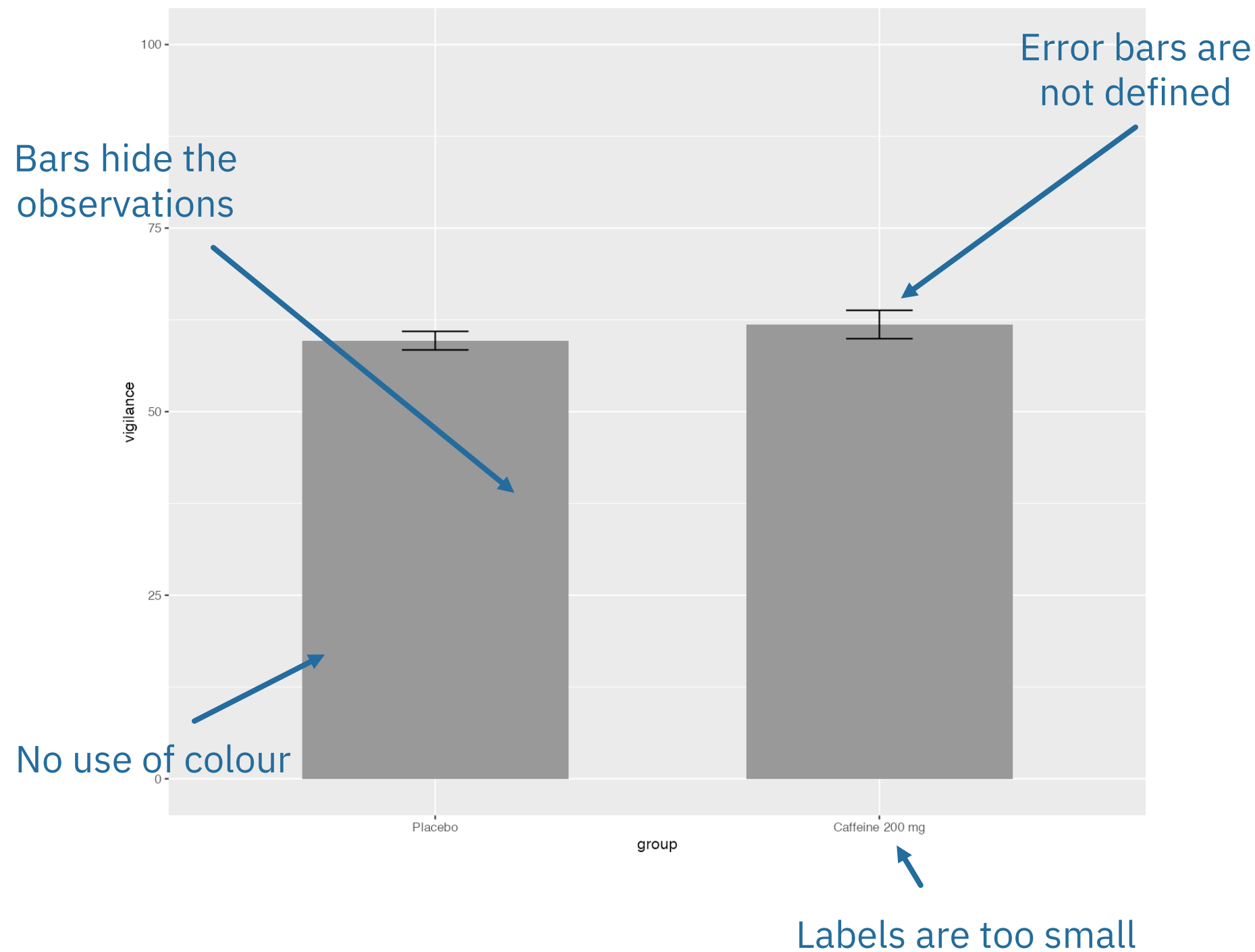


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The first draft – bar graph

What is this figure not telling you?



R code

```
p_first <- ggplot(summ,
  aes(group, mean)) +
  geom_col(fill = "grey60", width = 0.6) +
  geom_errorbar(aes(
    ymin = mean - se,
    ymax = mean + se), width = 0.15) +
  labs(x = "group", y = "vigilance") +
  scale_y_continuous(limits = c(0, 100)) +
  theme_grey(base_size = 7)
```

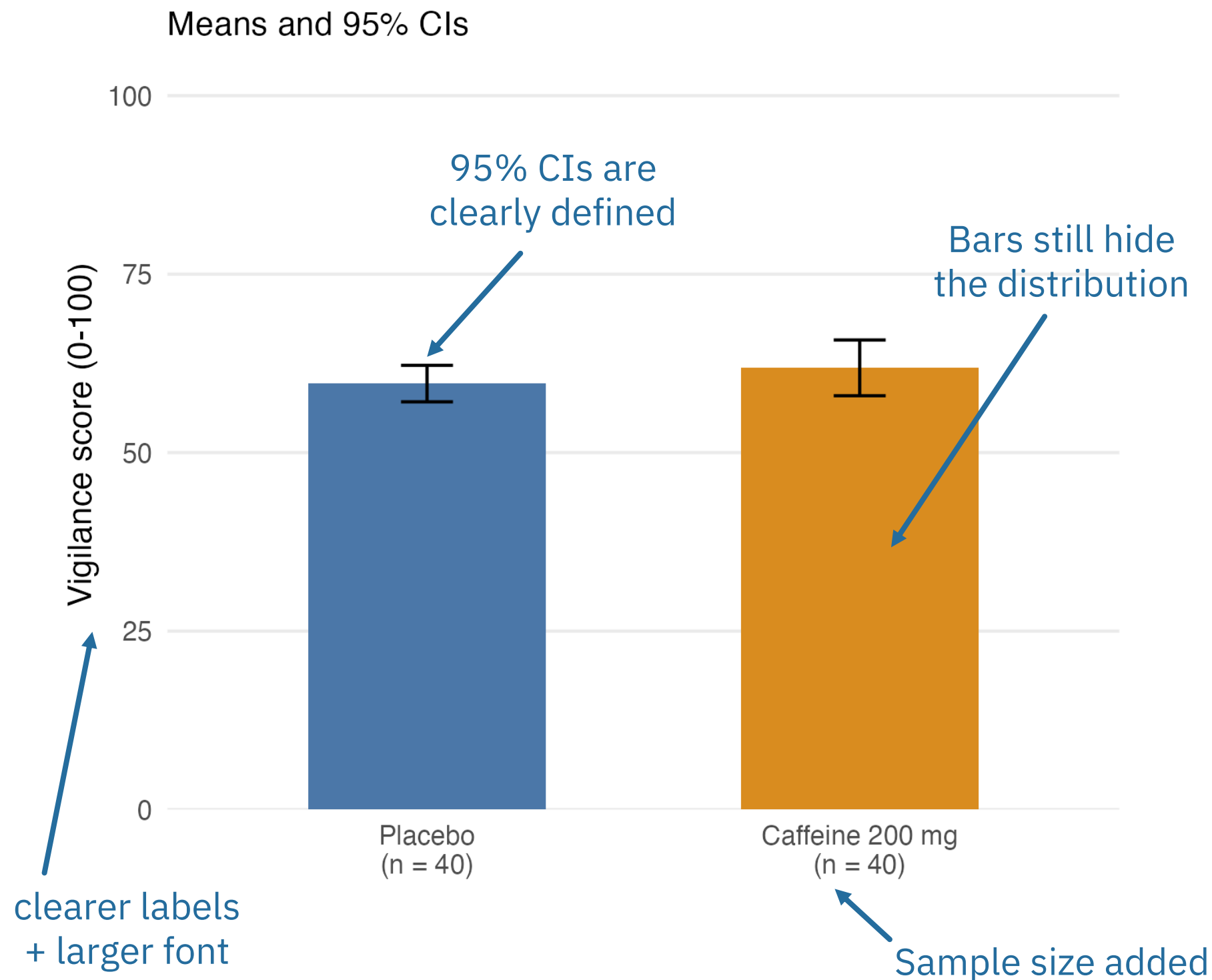


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The second draft

Clearer labels and uncertainty (but the observations are still hidden)



R code

```
p_bar <- ggplot(summ,
  aes(group, mean, fill = group)) +
  geom_col(width = 0.55) +
  geom_errorbar(aes(
    ymin = mean - ci,
    ymax = mean + ci), width = 0.12) +
  style +
  labs(subtitle = "Means and 95% CIs") +
  scale_y_continuous(
    limits = c(0, 100),
    expand = expansion(mult = c(0, 0.05)))
```

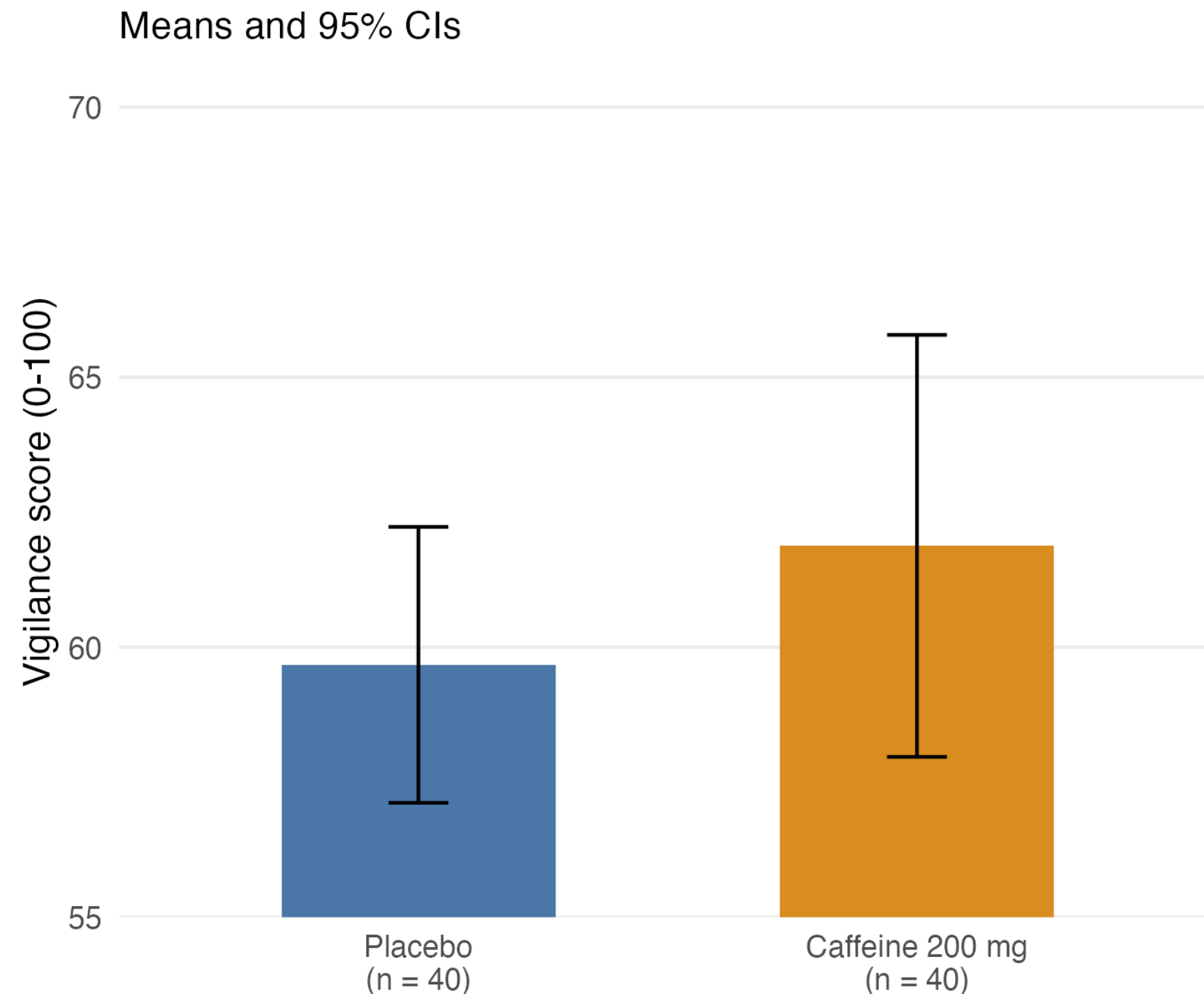


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A misleading bar-chart baseline (not recommended!)

The same group means look much further apart when the axis starts at 55

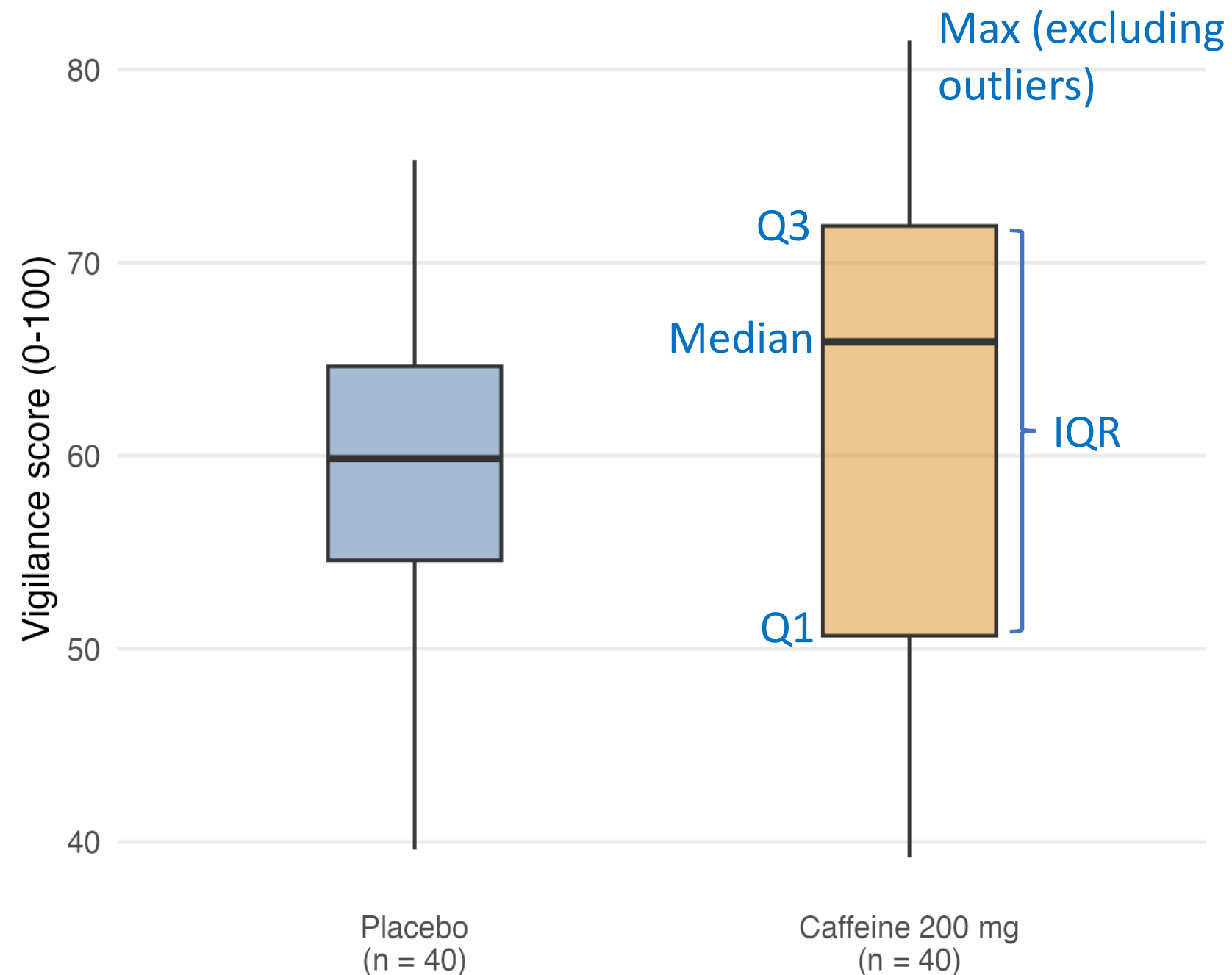


- Starting the axis at 55 *exaggerates differences*.
 - The caffeine bar looks almost 50% taller, although the group means differ by only about 2.2 points.
- Bar lengths should represent values from zero. Here, they represent only the amount above 55, giving a distorted impression of relative magnitude.
 - Recommend using a zero baseline, or a point-and-interval plot if a narrower axis is needed.*

Useful article:

Streit & Gehlenborg (2014). Bar charts and box plots
<https://www.nature.com/articles/nmeth.2807>

An alternative option - boxplots



R code

```
p_box <- ggplot(dat,  
  aes(group, vigilance, fill = group)) +  
  geom_boxplot(width = 0.35,  
    alpha = 0.5) +  
  style
```

- The median and middle 50% become visible
- The full range of data becomes clearer

BUT:

- Detailed shape of the distribution still not clear

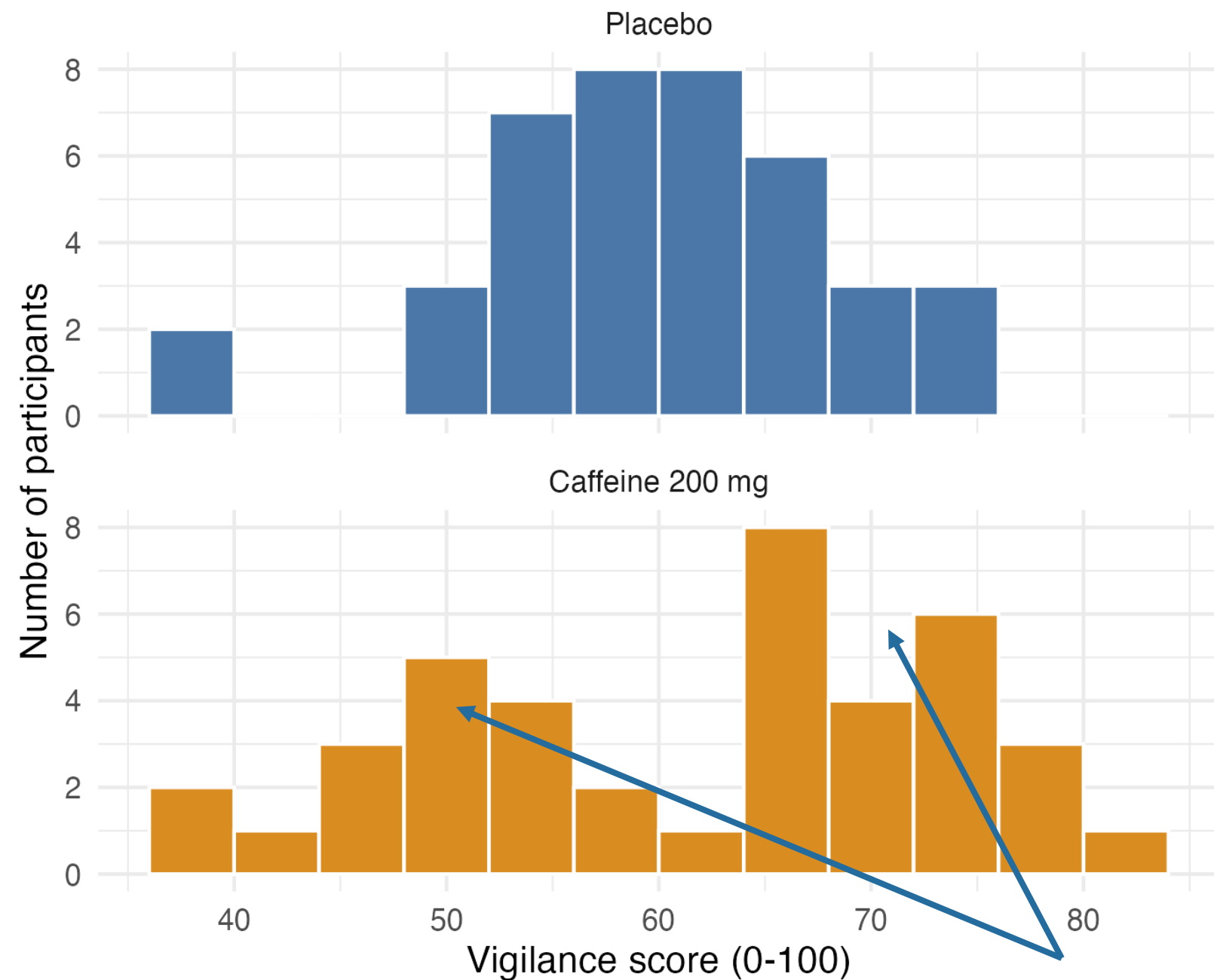


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Histograms

Histograms can sometimes be useful to reveal the shape of the distribution



Caffeine group appears to have a bi-modal distribution

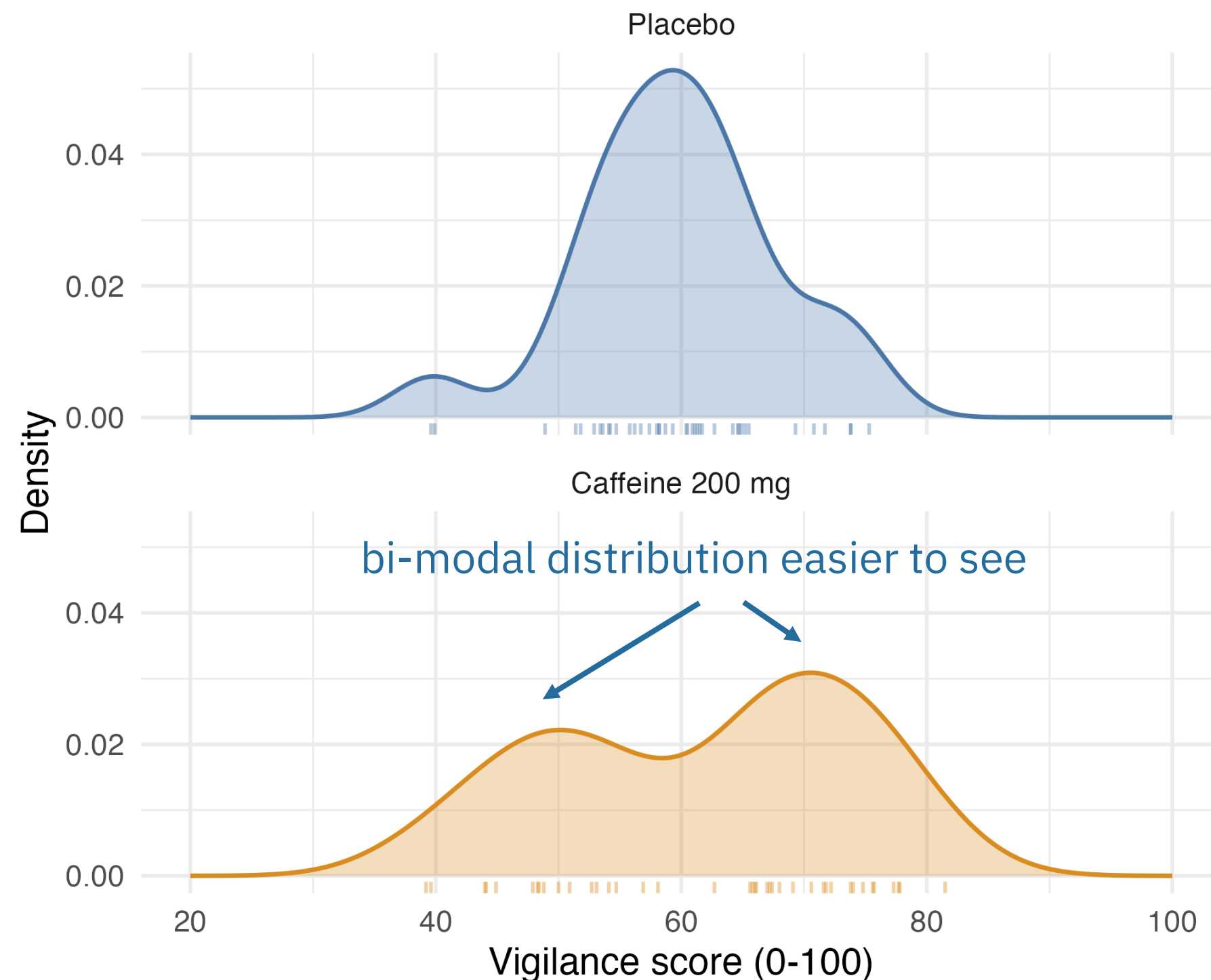
R code

```
p_hist <- ggplot(dat,  
  aes(vigilance, fill = group)) +  
  geom_histogram(binwidth = 4,  
    boundary = 0, colour = "white") +  
  facet_wrap(~ group, ncol = 1) +  
  scale_fill_manual(values = pal,  
    guide = "none") +  
  labs(x = "Vigilance score (0-100)",  
    y = "Number of participants") +  
  theme_minimal(base_size = 14)
```



This is more obvious if we use a density plot

A density plot gives us a smooth estimate of the distribution shape

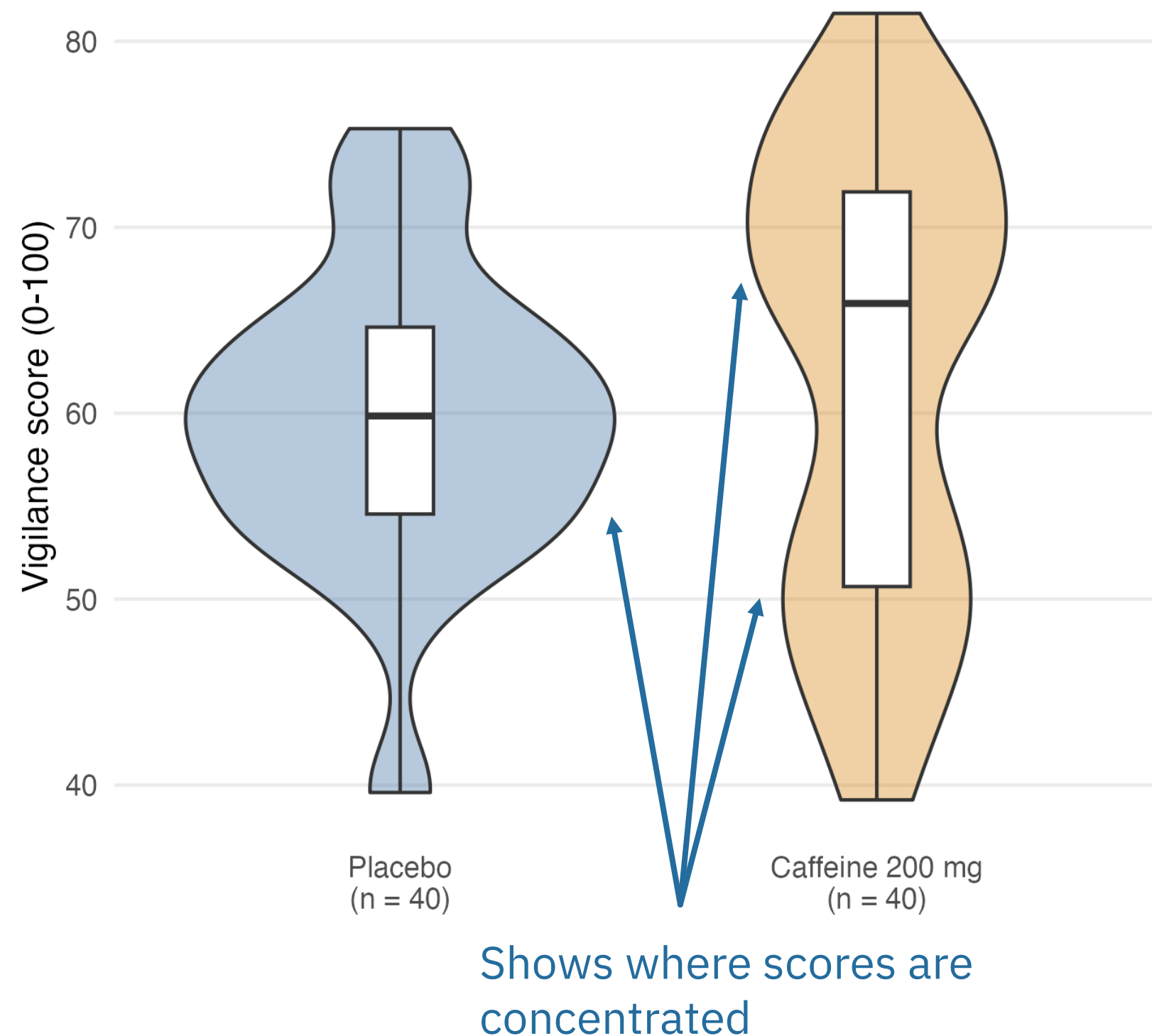


R code

```
p_density <- ggplot(dat,  
  aes(vigilance, fill = group,  
    colour = group)) +  
  geom_density(adjust = 1, alpha = 0.3,  
    linewidth = 0.8) +  
  geom_rug(alpha = 0.4, sides = "b") +  
  facet_wrap(~ group, ncol = 1) +  
  scale_x_continuous(limits = c(20, 100)) +  
  scale_fill_manual(values = pal,  
    guide = "none") +  
  scale_colour_manual(values = pal,  
    guide = "none") +  
  labs(x = "Vigilance score (0-100)",  
    y = "Density") +  
  theme_minimal(base_size = 14)
```



Violin plots



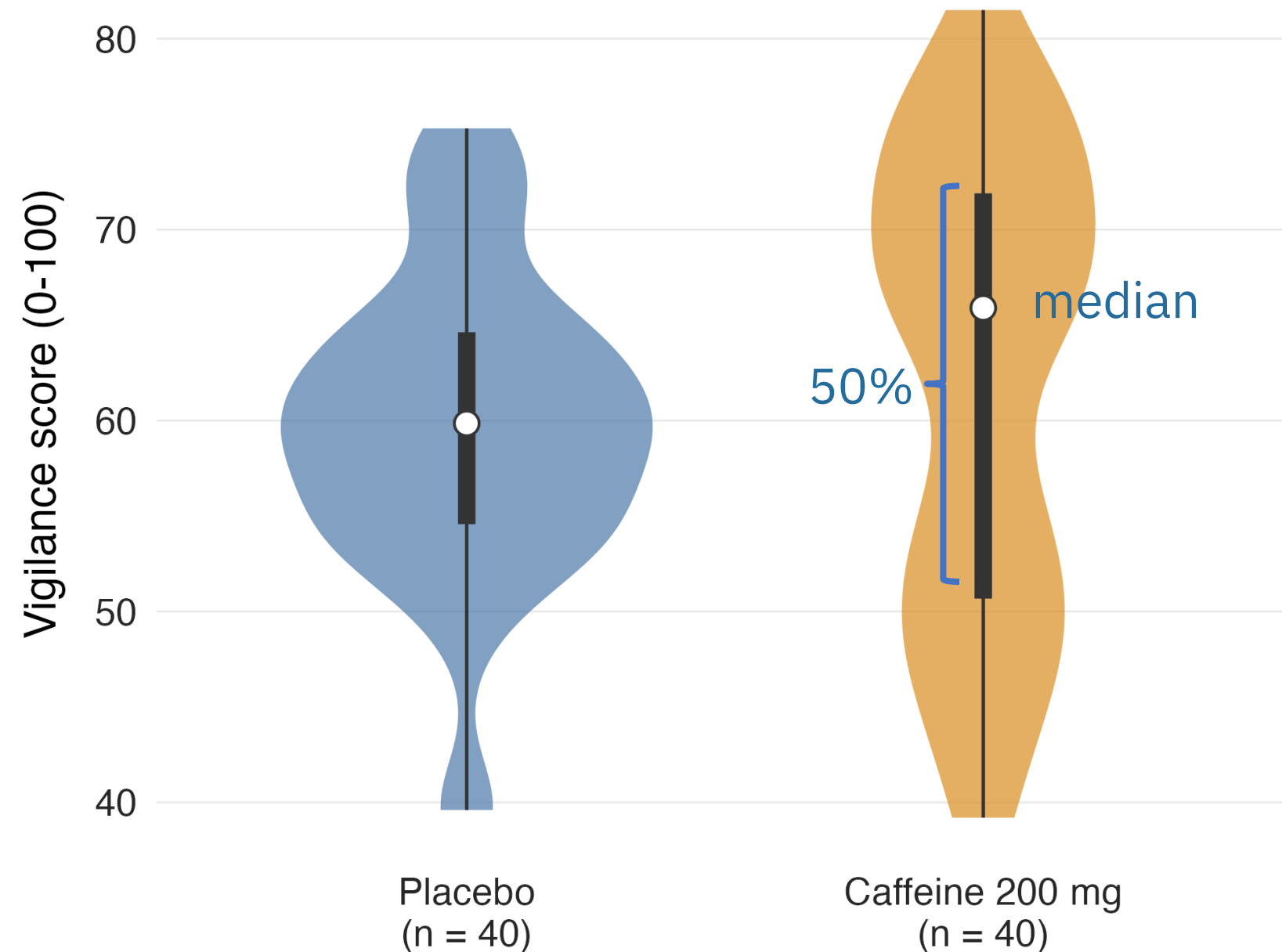
R code

```
p_violin <- ggplot(dat,  
  aes(group, vigilance, fill = group)) +  
  geom_violin(trim = TRUE,  
    adjust = 0.8, alpha = 0.4) +  
  geom_boxplot(width = 0.14,  
    fill = "white") +  
  style
```

- A violin plot adds the shape of the distribution to the boxplot
- Basically is a density plot flipped on its side + a boxplot!



A simpler violin option



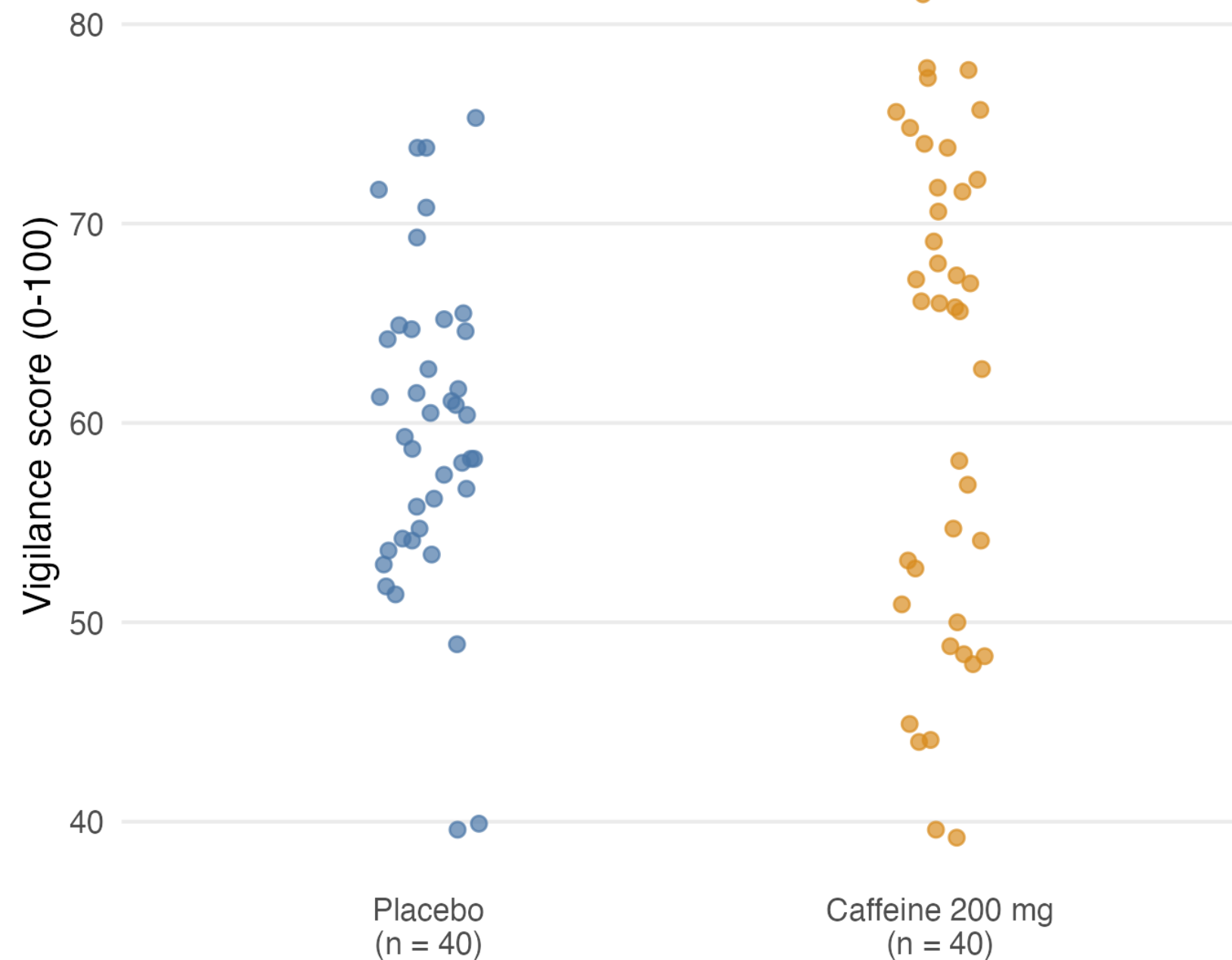
R code

```
p_compact <- ggplot(dat,
  aes(group, vigilance)) +
  geom_violin(aes(fill = group),
    width = 0.72, adjust = 0.8,
    alpha = 0.7, colour = NA) +
  geom_linerange(data = violin_summary,
    aes(y = centre, ymin = low, ymax = high),
    linewidth = 0.55, colour = "grey20") +
  geom_linerange(data = violin_summary,
    aes(y = centre, ymin = q1, ymax = q3),
    linewidth = 2.8, colour = "grey20") +
  geom_point(data = violin_summary,
    aes(y = centre), shape = 21, size = 3.5,
    fill = "white", colour = "grey20") +
  violin_style
```



What about individual scores?

This shows all individual observations, but can sometimes be difficult to read on its own.



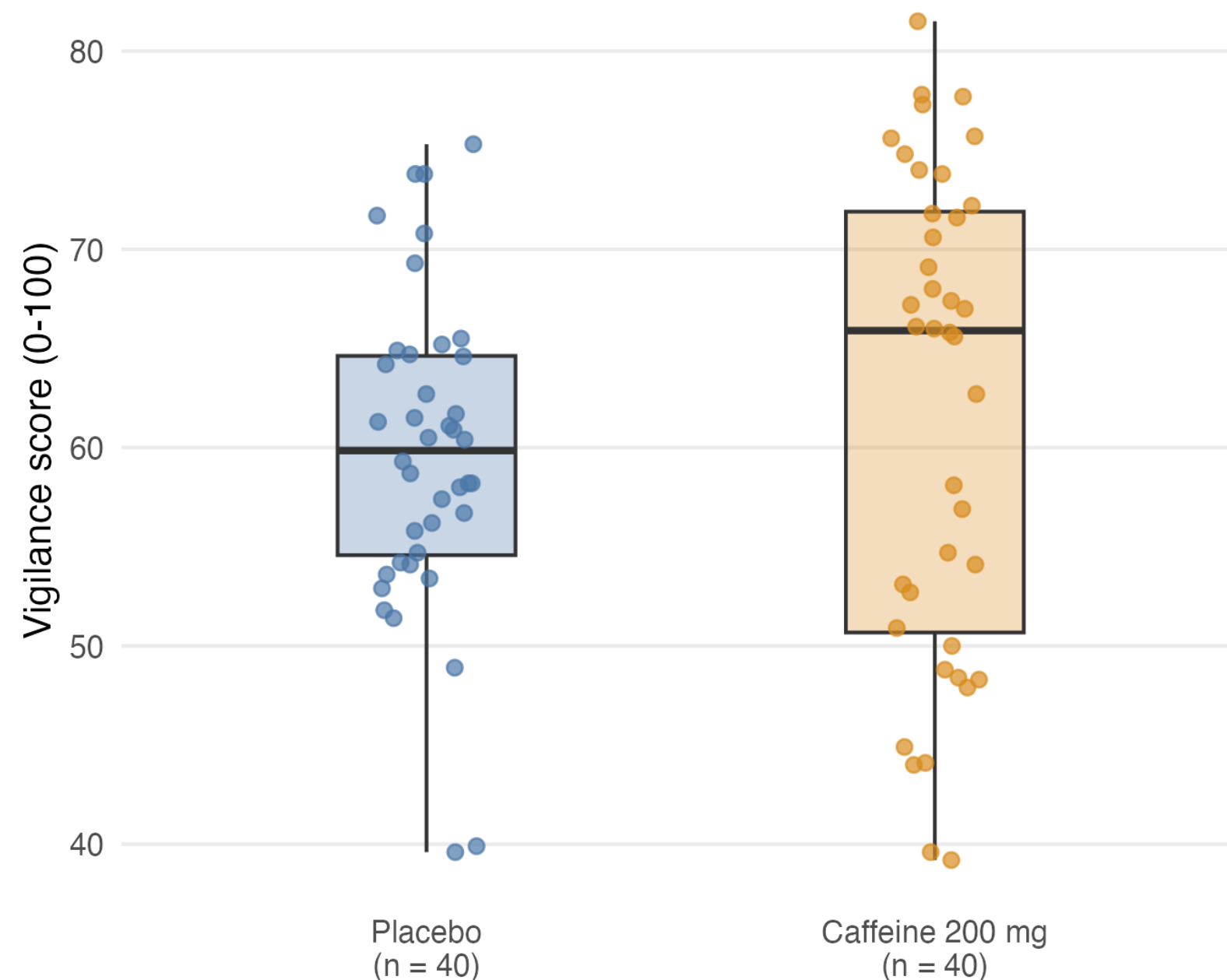
R code

```
p_jitter <- ggplot(dat,  
  aes(group, vigilance, colour = group)) +  
  geom_point(size = 2.2, alpha = 0.7,  
    position = position_jitter(  
      width = 0.10, height = 0,  
      seed = 1)) +  
  style
```



Points with a boxplot

In ggplot, we can easily overlay individual scores on a boxplot.



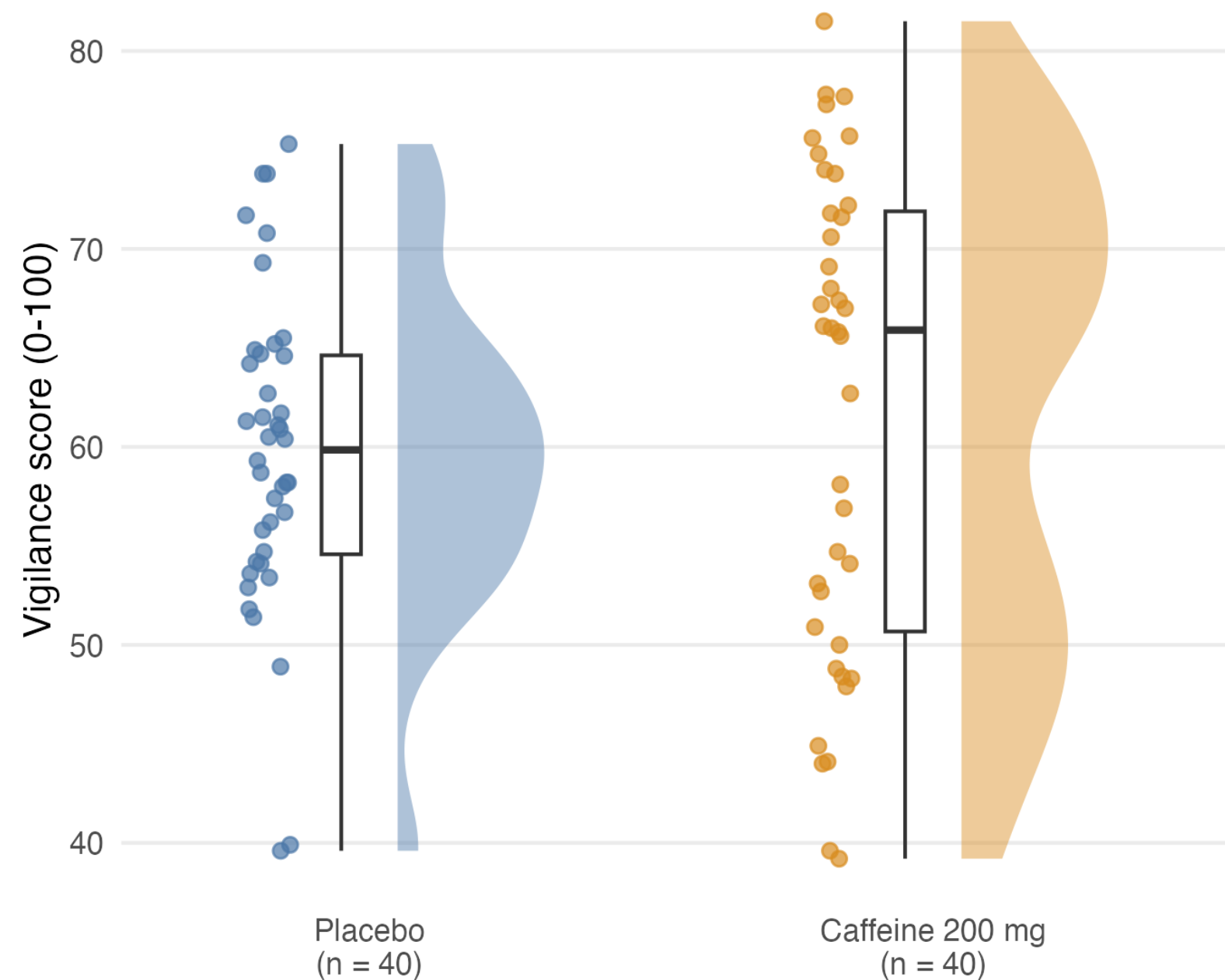
R code

```
p_boxpoints <- ggplot(dat,  
  aes(group, vigilance)) +  
  geom_boxplot(aes(fill = group),  
    width = 0.35, alpha = 0.3,  
    outlier.shape = NA) +  
  geom_point(aes(colour = group),  
    size = 2.2, alpha = 0.7,  
    position = position_jitter(  
      width = 0.10, height = 0, seed = 1)) +  
  style
```

Now we have all the individual data points as well as useful information from the boxplots such as median, data quartiles etc.



Raincloud plots



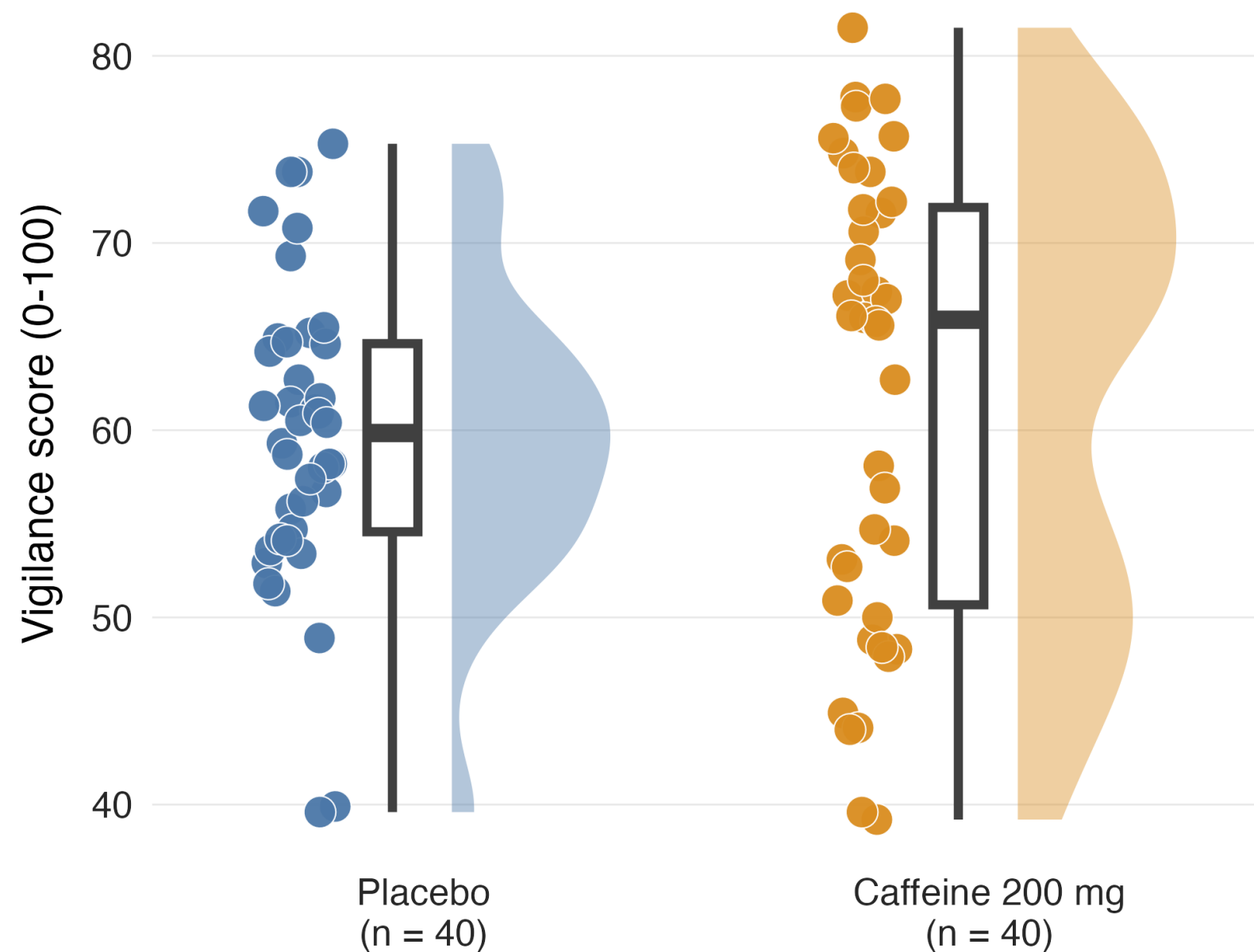
R code

```
p_rain <- ggplot() +  
  geom_polygon(data = cloud, aes(  
    xnum + 0.05 + width * 0.26, score,  
    fill = group), alpha = 0.45) +  
  geom_boxplot(data = dat, aes(  
    xnum - 0.05, vigilance, group = group),  
    width = 0.07, fill = "white",  
    outlier.shape = NA) +  
  geom_point(data = dat, aes(  
    xnum - 0.18, vigilance, colour = group),  
    size = 2.2, alpha = 0.7,  
    position = position_jitter(  
      width = 0.04, height = 0, seed = 1)) +  
  rain_style
```



Refining the raincloud

Larger labels and points, clearer boxes and a quieter background



R code

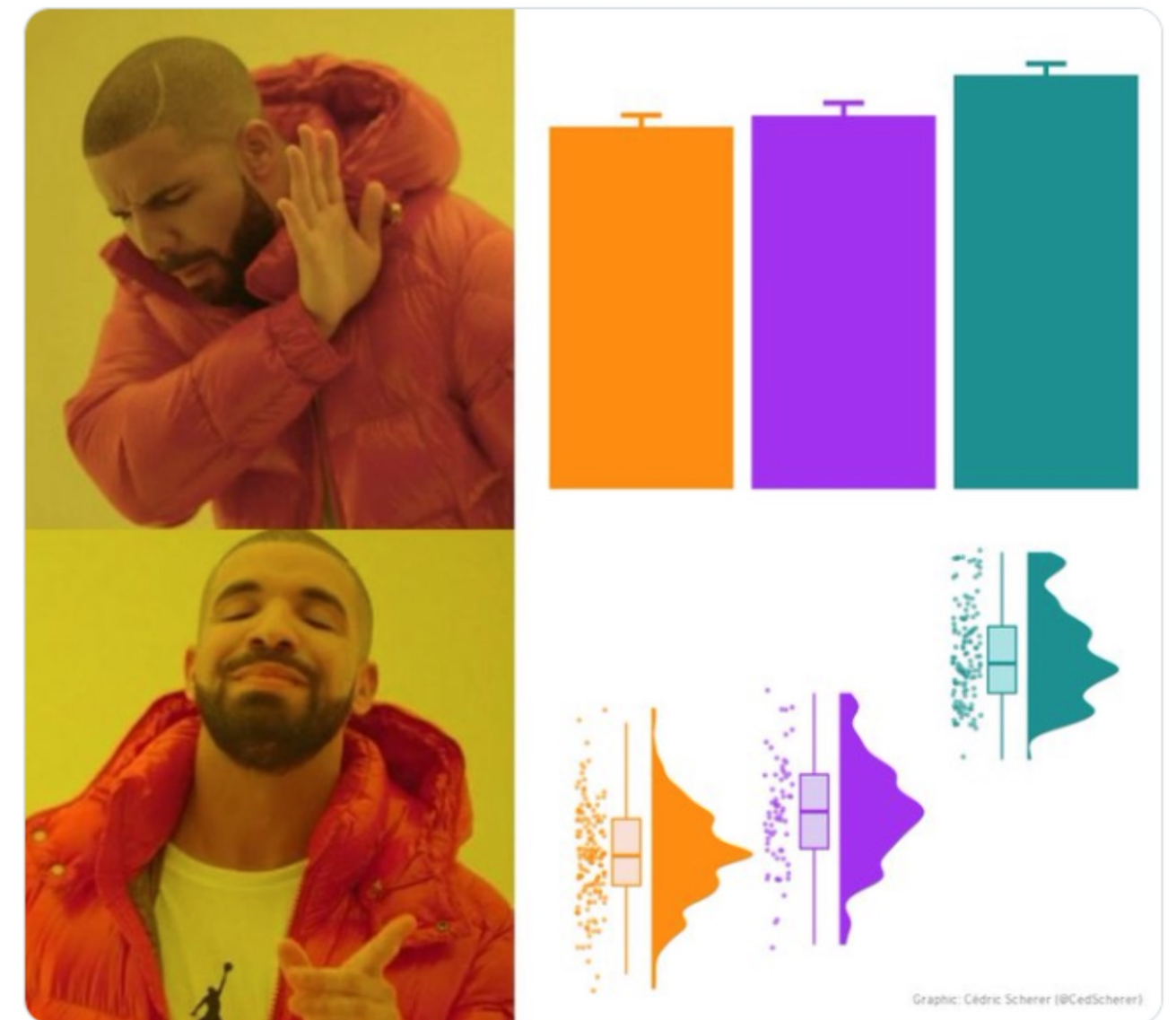
```
p_polished <- rain_canvas +  
  geom_boxplot(data = dat, aes(  
    xnum - 0.055, vigilance, group = group),  
    width = 0.09, fill = "white",  
    colour = "grey25", linewidth = 1.5,  
    outlier.shape = NA) +  
  geom_point(data = dat, aes(  
    xnum - 0.22, vigilance, fill = group),  
    shape = 21, size = 5, stroke = 0.35,  
    colour = "white", alpha = 0.95,  
    position = position_jitter(  
      width = 0.065, height = 0, seed = 1)) +  
  theme_paper
```

Figure X: Vigilance scores in the placebo and caffeine (200 mg) groups (n = 40 per group; simulated data). Points show individual participants, boxes show medians and interquartile ranges, and half-violins show the estimated distributions.



Questions for your next figure

- **Change one plotting choice at a time.** Try different colours, transparency, or amount of jitter.
- **Explain what readers are looking at.** Label the outcome and units, include sample sizes, and state what the points, lines and intervals represent.
- **Check the exported figure at its intended size.** Place it in the manuscript and check the labels, points and line widths (generally, I find fonts need to be a lot bigger than you initially think!).
- **Try another plot with the same data.** Compare your usual plot with an alternative, for example, replace a bar chart with points and a boxplot. Does it reveal anything the original figure hides?



Guide, code and data

Thanks for listening!

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